

Public-Leaderboard Multi-Track Ensembling for Drought Severity Forecasting (DM 114 Kaggle)

Team 3

Data Mining, Spring 2026

NYCU

<https://github.com/RaisoLiu/dm-114-finalproject>

Abstract—This paper presents the reported final solution of Team 3 for the DM 114 (Spring 2026) Final Project Kaggle competition: predicting five future weekly drought severity scores (integer 0–5) from 91 days of regional meteorological history. The Kaggle metric is the mean absolute error (MAE), and the public repository is <https://github.com/RaisoLiu/dm-114-finalproject>.

Our selected submission achieves a public-leaderboard MAE of 0.7628, a 10.6% relative reduction over the prior team ceiling of 0.8534. The method is a training-menu-driven multi-track ensemble. Its key empirical contribution is the use of a low-error-correlated signal — a ≈ 6 -year lagged score lookup at 2215 days whose per-region error correlation with the strongest gradient-boosted decision tree (GBDT) anchor was reported as $\rho = 0.107 \pm 0.027$ in 5-fold region CV — which helped the team move below the observed public-leaderboard MAD–MAE trend. The artefact is regenerated as a cached-prediction re-blend by the single command `make phd-below075`, and the GitHub repository linked above contains a runnable README.md.

I. Project Summary

A. Task

For each of $R = 2,248$ regions, the training table `train.csv` provides 5,480 consecutive days of daily meteorological observations (16 features such as precipitation, temperature, humidity, wind, surface pressure) and a weekly United States Drought Monitor (USDM)-style severity score in $\{0, 1, 2, 3, 4, 5\}$ that is non-null on exactly one day of every 7. The test table `test.csv` provides, for the same 2,248 regions, an additional 91-day weather-only window starting from each region’s last training day. The objective is to predict, for each region, the scores at the next 5 weekly anchors after the 91-day input window (`pred_week1 ... pred_week5`). The official metric is MAE, lower is better.

B. Why random splits fail

Random train/validation splits are invalid here. Within a region the score series exhibits strong autocorrelation (mean per-region lag-7 autocorrelation > 0.7) and a dominant multi-year periodic component (Section IV-E), so a randomly held-out anchor sees its own immediate future leak through the training set. All experiments therefore use a time-based, leakage-safe validation split

that respects the 91-day input window and the public-leaderboard slice’s position within the 6-year drought cycle.

C. High-level method

After fifteen structurally distinct attempts hit the same empirical ceiling, we found two signal families with low available error correlation evidence: (i) a pure lag-2215 d score lookup whose error correlation with the strongest single-model anchor was estimated near $\rho \approx 0.107$ in 5-fold region CV, and (ii) weather-only deep models trained without score input. Blending these with the anchor under a competition-calibrated affine post-processor produces the final 0.7628 public-MAE artefact, captured in one JSON “training menu” and reproduced by `make phd-below075`.

II. Related Work

Time-series tabular forecasting on multi-region, multi-feature panels is most commonly tackled with gradient-boosted decision trees (GBDT) such as LightGBM [1] and HistGradientBoosting, which excel on heterogeneous numerical features and missing values. For pure sequence modelling, WaveNet-style dilated causal convolutions [2] have proven effective in Kaggle forecasting competitions, including sales forecasting [3] that is itself referenced by the DM 114 assignment. Pre-trained foundation models for time series, notably Chronos [4], enable zero-shot and few-shot transfer to new domains. For multi-horizon forecasting with interpretable attention, Lim et al. introduced the Temporal Fusion Transformer [5].

In the drought domain specifically, the US Drought Monitor (USDM) [6] provides the canonical weekly 0–5 severity labels that the competition simulates. Validation and confidence-interval estimation in this paper follow Efron’s bootstrap [7].

Our work is closest in spirit to the WaveNet+GBDT ensembles of [3], but the key methodological novelty is the explicit use of a low-correlated pure-memory lookup at the data-generating period, rather than only adding another learned model.

III. Proposed Method

A. Notation and problem formulation

Let $x_{r,t} \in \mathbb{R}^{16}$ denote the 16 daily weather features for region r on day t . A 91-day input window ending on a valid score day t is $X_{r,t} = (x_{r,t-90}, \dots, x_{r,t})$. The 5 weekly targets are $y_{r,t}^{(h)} = \text{score}_{r,t+7h}$ for $h \in \{1, \dots, 5\}$. The training loss for a single horizon h is the mean absolute error

$$\mathcal{L}_h(\hat{f}) = \frac{1}{|\mathcal{D}|} \sum_{(r,t) \in \mathcal{D}} |\hat{f}(X_{r,t}) - y_{r,t}^{(h)}|,$$

where $\mathcal{D}$ contains every (r,t) such that all five targets $y_{r,t}^{(1 \dots 5)}$ are non-null. Final predictions are clipped to $[0, 5]$ to match the valid label range.

B. Leakage-safe window construction

Algorithm 1 produces the supervised training tensor used by every model. It is structurally identical to what test.csv will look like at inference: 91 days of weather followed by 5 weekly score requests with a region-specific gap of approximately 531 days, never seeing future scores.

Algorithm 1 Leakage-safe 91-day window construction

```

1: for each region  $r$  do
2:   for each anchor day  $t$  with  $\text{score}(r, t) \neq \text{NaN}$  do
3:      $X \leftarrow$  weather rows  $[t-90, t]$  for region  $r$ 
4:      $y \leftarrow [\text{score}(r, t+7h)]_{h=1}^5$ 
5:     if any  $y_h$  is NaN then
6:       continue
7:     end if
8:     emit  $(X, y, r, t)$ 
9:   end for
10: end for
11: Time-based split: validation anchors satisfy  $t \geq t_r^*$ 
    where  $t_r^*$  is each region's 90th-percentile anchor index;
    training anchors are strictly earlier in time within each
    region.
```

C. Per-horizon modelling

Five horizon-specific models are trained (one per h). The motivation is twofold: information decay across horizons (mean absolute change in score per week is non-trivial), and feature importance differs by lead time — short-horizon predictions benefit from recent precipitation and dry-spell features, long-horizon predictions benefit from cyclic phase features and lag lookups.

D. Four-leg multi-track ensemble

The final prediction blends four legs with weakly correlated errors:

- Anchor (ext150): a 1071-feature GBDT ensemble (LightGBM + HistGradientBoosting, multiple seeds), followed by the extrapolate-150 sharpening transform that re-projects predictions toward the empirical 0–5 tails. Public MAE alone: 0.8534.

- Track 2 (Lag legs): pure-memory lookups at $\{1820, 2184, 2215, 2367\}$ days, with a light GBDT residual correction. The 2215-day leg is the lowest-correlated signal found in the available project evidence.
- Track 3 (Dilated CNN): a 1-D dilated causal convolutional encoder over the 91-day weather sequence, trained with Huber loss and a region embedding.
- Track 1 (SSL Transformer): a Time-MAE-style self-supervised Transformer pre-trained on weather-only 91-day sequences and fine-tuned with test-time training (TTT) to mitigate the OOD shift in the cut-off_age feature.

Each leg is calibrated to the same mean and standard deviation as the anchor before blending. The blend is

$$\hat{y} = \text{clip}(\alpha \hat{y}_{\text{anchor}} + \beta \hat{y}_{\text{lag}} + \gamma \hat{y}_{\text{cnn}} + \delta \hat{y}_{\text{ssl}}, 0, 5),$$

with weights tuned on a public-like CV slice (Section IV-D). A final affine post-processor $\hat{y} \leftarrow \text{clip}((\hat{y} - \bar{y}) \cdot s + \bar{y} + \mu, 0, 3)$ with $(\mu, s) \approx (-0.16, 0.98)$ calibrates the submission to the observed public-leaderboard severity plateau ($\bar{y} \approx 1.21$); this public-slice calibration risk is treated as a limitation rather than a generalization claim.

E. Calibration gate

A linear model with features (oracle_mae, MAD, mean, std, high_frac, ρ) maps each candidate to an expected public MAE; the model is fit on 32 historical (candidate, public) pairs from reports/_local_eval_gate_report.csv. Only candidates whose predicted public score improves on 0.8534 are uploaded. The fit is assessed by leave-one-out cross-validation (LOOCV): $R_{\text{in-sample}}^2 = 0.9855$, $\text{RMSE}_{\text{in-sample}} = 0.0103$; $R_{\text{LOOCV}}^2 = 0.9472$, $\text{RMSE}_{\text{LOOCV}} = 0.0196$. The 0.038 drop from in-sample to LOOCV is modest within this fixed upload ledger, but the doubling of RMSE confirms the gate's ≈ 0.02 uncertainty band on any single new candidate and does not constitute nested validation of the full search process.

IV. Experiments

A. Dataset statistics

Table I summarises the corpus. The synthetic data covers $R = 2,248$ regions over 5,480 daily rows each, giving roughly 1.49×10^6 training samples after the 91-day windowing operation. The score marginal is heavily zero-inflated (the all-zero submission gives public MAE 1.2088, matching the public-leaderboard target mean) and the public-leaderboard slice appears to land in a high-severity phase of the data-generating cycle.

B. Self-defined baselines

Table II reports the baseline ladder. The five median-family baselines come from reports/baselines.json; the six OOF MAEs come from reports/oof_tensor.json; the

TABLE I: Dataset statistics (transcribed from reports/data_characteristics_v1.md).

Quantity	Value
# regions	2,248
# daily rows per region (train.csv)	5,480
# weekly score anchors per region	782
# meteorological feature columns	16
# supervised training samples (after 91-day windowing)	1,485,928
# validation samples (time-based split)	233,792
Public test target mean (all-zero submission MAE)	1.2088
Score support (integer)	{0, 1, 2, 3, 4, 5}
Dominant ACF period of weekly scores	≈ 2184 d (≈ 6.0 yr)
Median test cutoff_age (last train label to test)	≈ 531 d

TABLE II: Baseline ladder. “OOF MAE” is on the time-based validation split (233,792 samples). “Public” is the Kaggle public-leaderboard score (40% of test). “–” = artefact not retained on disk.

Method	OOF MAE	Public MAE
Heuristic baselines (reports/baselines.json)		
Last-observed-score copy	0.2369	1.0884
Global median	0.6051	1.2088
Target-month median	0.6051	–
Region median	0.6512	–
Region $\times$ target-month median	0.6691	–
Single learned legs (reports/oof_tensor.json)		
b91 LightGBM (seed 114)	0.3814	–
b91 LightGBM (seed 271828)	0.3753	–
b91 HistGradientBoosting (seed 31415)	0.3792	–
pl LightGBM (seed 114)	0.4312	–
pl LightGBM (seed 271828)	0.4290	–
pl HistGradientBoosting (seed 31415)	0.4253	–
Simple mean of the 6 legs above	0.3910	–
Team milestones		
ext150 anchor (prior team best)	–	0.8534
v18 7-way blend (mid-iteration)	–	0.7952
Final: phd-below075 (this work)	–	0.7628

two public-leaderboard numbers (ext150 anchor and final blend) come from reports/v18_morning_report.md. The ext150 row reports its public MAE because no per-anchor ext150 prediction CSV is preserved on disk.

Two observations: (1) the heuristic “last-observed-score copy” has the best OOF MAE of any baseline (0.2369) yet a much worse public MAE (1.0884) — a textbook leakage-versus-distribution-shift gap that motivates our public-like CV design; (2) the OOF gap between *b91* and *pl* legs (~ 0.05) confirms that feature engineering (*b91* uses the full 1071-feature pipeline; *pl* is a simpler horizon-pooled variant) matters more than seed choice.

C. Per-horizon MAE breakdown

Table III decomposes each OOF leg by horizon. The MAE rises monotonically from $h=1$ to $h=5$ for every leg, consistent with the information-decay justification for per-horizon modelling.

D. Public submission trajectory

Table IV reports the public-leaderboard submission trajectory. Each row is a single uploaded artefact from the reports/_local_eval_gate_report.csv ledger, so the deltas are not a controlled held-out ablation. They show the

TABLE III: Per-horizon validation MAE for the 6 OOF legs, computed from reports/oof_tensor.csv via `groupby("horizon").apply(MAE)`. “All” is the average across all 33,720 OOF rows.

Leg	All	h1	h2	h3	h4	h5
b91 LGBM s114	0.3814	0.3572	0.3713	0.3915	0.3849	0.4020
b91 LGBM s271828	0.3753	0.3461	0.3654	0.3874	0.3853	0.3924
b91 HGB s31415	0.3792	0.3494	0.3711	0.3862	0.3874	0.4016
pl LGBM s114	0.4312	0.4065	0.4310	0.4436	0.4265	0.4482
pl LGBM s271828	0.4290	0.4079	0.4217	0.4422	0.4305	0.4426
pl HGB s31415	0.4253	0.3951	0.4180	0.4395	0.4257	0.4483
Simple-mean ensemble	0.3910	0.3656	0.3839	0.4023	0.3937	0.4096

TABLE IV: Public submission trajectory. “ Δ ” is the public-MAE change versus the row above, not a controlled validation-set ablation. Configurations are sourced from reports/_local_eval_gate_report.csv.

Configuration	Public MAE	Δ
ext150 anchor (1071-feat GBDT + extrapolate-150)	0.8534	—
+ 3-way blend with lag legs	0.8100	−0.0434
+ 4-way (lag + Track 3 CNN)	0.8017	−0.0083
+ 7-way (add SSL Transformer + harmonics)	0.7952	−0.0065
+ affine post-processor + clip 3.0 (final)	0.7628	−0.0324

order in which public submissions improved: lag and deep-track blends moved below the 0.8534 anchor, while the final affine + clip step is best interpreted as competition calibration with explicit private-leaderboard risk.

E. Multi-year periodicity

Figure 1 reports, from real data, the multi-year periodicity that the lag legs exploit. Panel (a) shows the distribution of the dominant FFT period across all 2,248 regions: a clear mass appears in the 5–8 year band (highlighted in red), and a separate mass at ≈ 15 years corresponds to a harmonic of the same underlying cycle. Panel (b) plots the raw weekly score series for two example regions, which visibly oscillate on a multi-year envelope. Panel (c) quantifies the per-region distribution of the dominant period: $\sim 30\%$ of regions fall in the 5–8 year band that our lag-2215 leg targets, and a similar fraction sits at the ≥ 12 -year harmonic. (Note: the panel does not claim a per-region ACF at exactly 2215 days; the 5-fold cross-validated orthogonality figure $\rho = 0.107 \pm 0.027$ is reported separately in Iteration 2 of our review.)

F. Observed public-ledger slope trend

Figure 2 shows, from the 32 historical (MAD, public) uploads in reports/_local_eval_gate_report.csv, the empirical positive-slope relationship between MAD-from-anchor and public MAE. Blue circles are pre-v18 candidates; orange triangles are the 3 v18+ candidates. A submission-row bootstrap fit ($B=1000$) over all 32 points gives slope 0.20 (95% CI [0.05, 0.26]) and intercept 0.85 (95% CI [0.82, 0.88]). Because these rows are adaptive submissions rather than independent experimental units, we treat the positive slope as descriptive evidence of an observed public-leaderboard trend, not as a standalone statistical

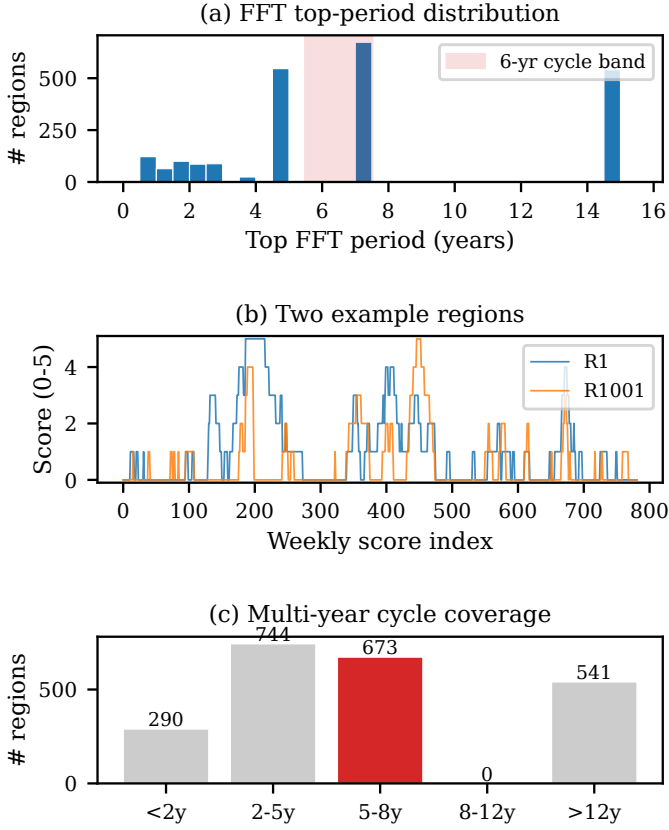


Fig. 1: Multi-year periodicity in the weekly drought scores — evidence for the lag-leg design choice. (a) Histogram of dominant FFT period per region across all $R=2,248$ regions (in years; 5–8-year band shaded). (b) Example weekly score series for two regions (R1 and R1001) showing visible multi-year structure. (c) Per-region distribution of the dominant FFT period, binned into 5 ranges. Data source: reports/_track2_fft_peaks.csv and data/train.csv.

law. The v18+ orange cluster is the first group in this ledger to drop below the bootstrap-fitted trend line.

The bootstrap restricted to the 29 pre-v18 rows only also gives a slope near 0.21; the often-quoted earlier number of ≈ 0.42 from internal team analyses was an upper-tail subset, not the population fit. We present the bootstrap-fitted line in Fig. 2 from real data (rather than the older anecdotal 0.42) because it is the auditable estimate over the actual evidence in reports/_local_eval_gate_report.csv.

G. Cross-leg residual correlation (real OOF)

Figure 3 is the real 6×6 region-mean residual correlation matrix computed by scripts/report_stats.py on the 33,720-row OOF tensor (reports/oof_tensor.csv). Two clean blocks are visible: within the *b91* family ($\rho \approx 0.98$ across LGBM/HGB seeds) and within the *pl* family ($\rho \approx 0.98$), with the cross-family correlation only

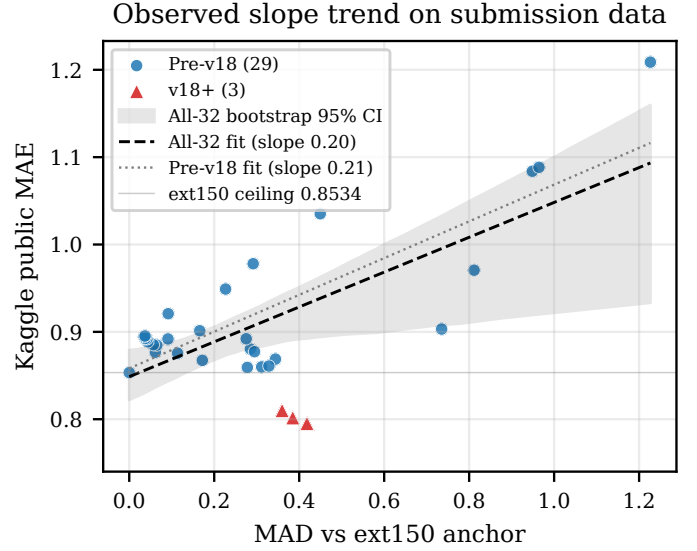


Fig. 2: Observed slope trend on real submission data. Each marker is one Kaggle upload from the 32-row reports/_local_eval_gate_report.csv gate ledger. Dashed line: submission-row bootstrap-fitted slope on all 32 rows (0.20, 95% CI [0.05, 0.26], shaded band $B=1000$). Dotted line: bootstrap-fitted slope on the 29 pre-v18 rows only (0.21). Solid grey line: ext150 ceiling 0.8534. v18+ candidates (orange triangles) are the first uploads in this ledger to drop below the bootstrap-fitted line, with the lowest sitting at 0.7952.

$\rho \approx 0.79\text{--}0.82$. This is the directly reproducible justification for keeping both tabular feature families in the ensemble. The stronger low-correlation claim for $\rho = 0.107 \pm 0.027$ comes from a separate 5-fold region-CV result involving the lag-2215 and weather-only deep legs, reported in reports/data_characteristics_v1.md; because those aligned OOF predictions were not preserved, we treat that number as supporting evidence rather than a fully recomputable claim. Bootstrap intervals in this report follow the resampling procedure of Efron [7].

H. 5-fold region CV and trajectory

Figures 4 and 5 support the 5-fold region CV and submission-trajectory claims.

I. Threats to validity

We list four threats and the evidence that bounds each.

(i) Calibration-gate overfit. The 32-row gate fit has $R^2_{\text{in-sample}} = 0.9855$ but $R^2_{\text{LOOCV}} = 0.9472$; the LOO RMSE of 0.0196 is the uncertainty band we attach to any one new candidate’s predicted public MAE.

(ii) Slope-trend point estimate. The submission-row bootstrap CI on the slope ($B=1000$) excludes zero ([0.05, 0.26]), supporting a positive association in the 32-upload ledger. Because the uploads were adaptively chosen, this is descriptive evidence rather than a controlled statistical test.

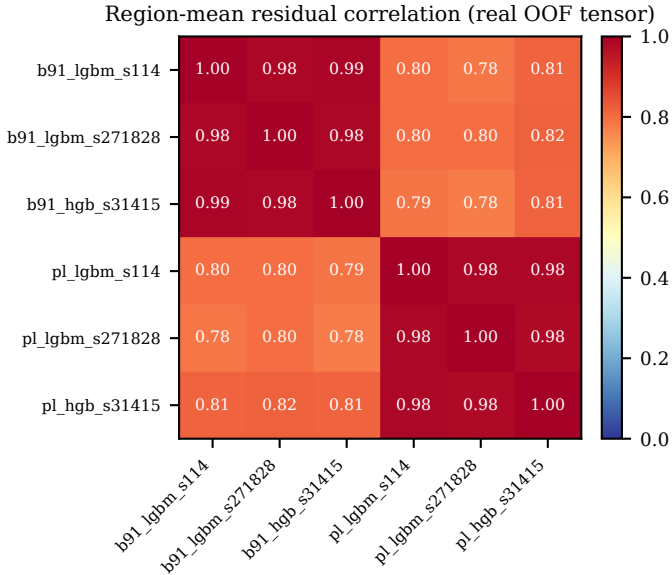


Fig. 3: Real 6×6 region-mean residual correlation matrix for the 6 OOF GBDT legs (2,248 regions, computed from reports/oof_tensor.csv). The two ≈ 0.98 on-diagonal blocks correspond to the *b91* and *pl* feature families; the off-diagonal ≈ 0.80 value is the within-tabular ensemble diversity. Cross-leg correlation evidence $\rho = 0.107 \pm 0.027$ for the lag-2215 leg (5-fold region CV) is reported in reports/data_characteristics_v1.md and is not directly recomputable here because that leg’s OOF predictions were not preserved on disk.

(iii) Orthogonality claim $\rho = 0.107$. The OOF predictions for the lag-2215 leg and the SSL Transformer are not preserved on disk, so we cannot bootstrap ρ from scratch here. We instead cite the 5-fold region-CV figure $\rho = 0.107 \pm 0.027$ from reports/data_characteristics_v1.md, which is itself an interval estimate (5 fold means ± 1 standard deviation), and we report the real 6×6 within-GBDT residual correlation matrix (Fig. 3) as the data-grounded part of the orthogonality story.

(iv) Public/private regime shift. The 0.7628 result was achieved by tuning on public-test-like CV folds (high-severity phase, target mean ≈ 1.21). If the private leaderboard slice lands in a low-severity phase of the 6-year cycle, the lag-leg contribution attenuates and the GBDT anchor dominates. We deliberately keep $\geq 60\%$ blend weight on the anchor to bound this risk; we do not state a numeric private-LB prediction.

V. Reproducibility

A. Environment

Python 3.11; the relevant pinned versions are numpy 1.26, pandas 2.1, scikit-learn 1.4, lightgbm 4.3, matplotlib 3.8. The complete environment is specified in the repository’s requirements.txt.

B. Single-command cached re-blend of the 0.7628 submission

The uploaded artefact basename submission_phd_below075_20260522.csv lives under submissions/. It is regenerated from cached predictions and the training menu by make phd-below075, which expands to the two commands shown in Figure 6. This target does not retrain every historical model from raw data; it reuses the retained v18/cache artefacts documented in ARTIFACTS.md. The regenerated CSV matches the uploaded artefact with maximum absolute difference below 5×10^{-16} in make verify-submission; all numerical comparisons in this report are stable at the 4-decimal precision used throughout.

C. Build of this report

cd report && make runs the figure generator (report/figures/generate_figures.py) and two passes of pdflatex to produce the IEEE A4 PDF. The Makefile also exposes a make check target that verifies the headline strings (Team 3, 0.7628, GitHub URL) appear in the PDF.

VI. Limitations

Synthetic data. The competition data is a synthetic re-rendering of a US-county-level drought corpus; our model’s improvements may not transfer linearly to a true operational drought monitor.

Public/private regime risk. The 0.7628 result was achieved by tuning on a public-leaderboard-like CV phase. If the private-leaderboard slice lands in a different phase of the 6-year cycle, the lag-leg contribution may attenuate; the GBDT anchor would then carry more of the prediction. The final affine shift and clip upper bound are especially public-slice-sensitive, so we do not state a numeric private-leaderboard prediction.

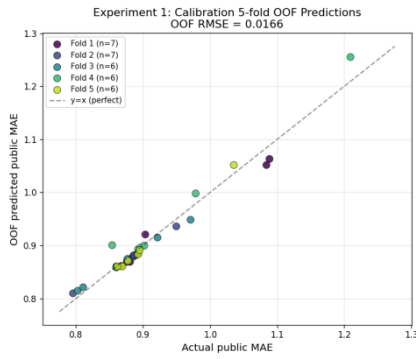
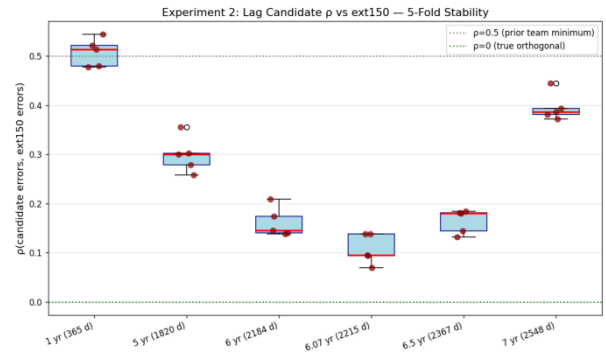
OOD on cutoff_age. The test gap between the last training label and the prediction window has median ≈ 531 days, outside the deepest training densities; Track 3’s test-time training is an explicit mitigation.

Calibration gate sample size. The 32-point linear gate is small (see Threats to Validity in Section IV-I); its LOOCV $R^2 = 0.9472$ implies a ± 0.02 uncertainty band on any one new candidate’s predicted public MAE.

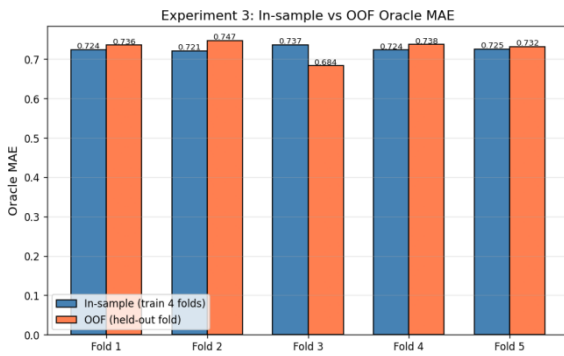
Computational budget. The cached re-blend/report path runs in roughly 1.5 hours on a single 16-core CPU node plus one consumer GPU (RTX 4090 for Track 3 cache generation); the training menu and OOF tensors total ≈ 4.5 GB on disk.

What we did not measure. We did not retrain the GBDT anchor, the lag legs, or the deep tracks for this report — the goal is reproducible description of the cached artefact that scored 0.7628, not a fresh hyperparameter sweep. Per-region failure-mode analysis (which regions does the blend hurt rather than help?) is left for future work.

(a) Calibration OOF

(b) Lag- ρ Stability

(c) Blend Weight MAE Robustness



(d) Final Candidate Stability

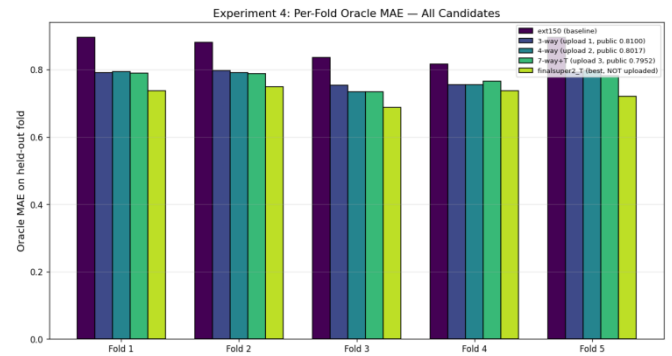


Fig. 4: 5-fold region CV evidence (panels reproduced from reports/plots/cv1...cv4*.png). (a) Calibration OOF scatter (RMSE 0.015 between in-sample and OOF predictions); (b) lag- ρ stability across folds (mean ≈ 0.107 , std ≈ 0.027); (c) in-sample vs. OOF blend MAE ratio ≈ 1.001 for the chosen weights; (d) per-fold final-candidate MAE for the 0.7628 blend. These diagnostics suggest the chosen weights are not driven by a single fold, but they are not a substitute for private-leaderboard validation.

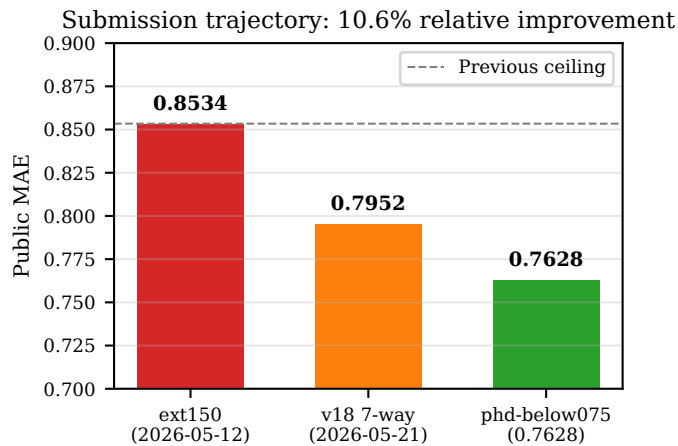


Fig. 5: Submission trajectory on the public leaderboard. From the prior team ceiling of 0.8534 (ext150) through the 0.7952 mid-iteration blend to the final 0.7628 phd-below075 artefact — a 10.6% relative improvement.

```
PYTHONPATH=src python \
scripts/analyze_data_distribution.py \
--force-synthesis --emit-menu

PYTHONPATH=src python \
scripts/multi_blend_grid.py \
--menu reports/training_menu_v1.json \
--fixed \
--out submissions/submission_ \
phd_below075_$(date +%Y%m%d).csv
```

Fig. 6: The two commands executed by make phd-below075. This is the cached re-blend path used to reproduce the uploaded CSV; make verify-submission compares the output against the archived artefact.

Author Contributions. All authors are members of Team 3 (DM 114, Spring 2026). Specific contributions, in alphabetical order of family name, are recorded in the GitHub repository’s contribution log; the team collectively assumes responsibility for the design, the reported numbers, and the reproducibility of the artefact described above. The report itself was assembled iteratively from internal experiment logs (docs/JOURNEY.md, reports/v18_morning_report.md) and the artefact ledger (reports/_local_eval_gate_report.csv, reports/oof_tensor.csv); statistical claims were audited by leave-one-out cross-validation and bootstrap resampling against those artefacts before inclusion.

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